

Tut - 12

```

1. #include <stdio.h>

int ways (int n) {
    int ans;
    if (n <= 1) {
        return 1;
    }
    ans = ways (n-1) + ways (n-2);
    return ans;
}

int main () {
    printf ("%d", ways (5));
    return 0;
}

```

```

2. #include <stdio.h>
#include <string.h>
void swap (char* a, char* b) {
    char temp = *a;
    *a = *b;
    *b = temp;
}

void print Permutations (char* arr, int left, int right)
{
    if (left == right) {
        printf ("%s\n", arr);
    } else {
        for (int i = left; i <= right; i++) {
            swap (&arr [left], &arr [i]);
            print Permutations (arr, left + 1, right);
            swap (&arr [left], &arr [i]);
        }
    }
}

```

```
void printIntegerPermutations (int *arr, int left, int right)
{
    if (left == right) {
        for (int i = 0; i <= right; i++) {
            printf ("%d", arr[i]);
        }
        printf ("\n");
    } else {
        for (int i = left; i <= right; i++) {
            int temp = arr[left];
            arr[left] = arr[i];
            arr[i] = temp;
            printIntegerPermutations (arr, left + 1, right);
            arr[i] = arr[left];
            arr[left] = temp;
        }
    }
}
```

```
int main () {
    int choice;
    printf ("Enter 1 for String or 2 for Integer Array: ");
    scanf ("%d", &choice);
```

```
    if (choice == 1) {
        char str[100];
        printf ("Enter a string: ");
        scanf ("%s", str);
        int n = strlen(str);
        printPermutations (str, 0, n-1);
    } else if (choice == 2) {
        int n;
```

```
        printf ("Enter the number of elements in the array: ");
```

```
scanf("%d", &n);
int arr[n];
printf("Enter the number of elements in the array: ");
scanf("%d", &n);
int arr[n];
printf("Enter the element, of the array: ");
for (int i = 0; i < n; i++) {
    scanf("%d", &arr[i]);
}
PrintIntegerPermutations(arr, 0, n-1);
} else {
    printf("Invalid choice %d", n);
}
return 0;
}
```

③

```
void printSubsetSums (int arr[], int n, int index, int sum) {
    if (index == n) {
        printf("%d\n", sum);
        return;
    }
    printSubsetSums (arr, n, index + 1, sum);
    printSubsetSums (arr, n, index + 1, sum + arr[index]);
}
```

```
int main () {
    int n;
    printf("Enter the number of elements in the array: ");
    scanf("%d", &n);
    int arr[n];
```

```

printf("Enter the elements of the array: ");
for (int i = 0; i < n; i++) {
    scanf("%d", &arr[i]);
}

```

```

printf("Sum of all subset: ");
printSubsetSum(arr, n, 0, 0);
return 0;
}

```

```

2) void printEvenOdd (int start, int end) {
    if (start > end) return;
    if (start % 2 == 0)
        printf("%d is even\n", start);
    else
        printf("%d is odd\n", start);
    printEvenOdd (start + 1, end);
}

```

```

int main () {
    int start, end;
    printf("EvenOdd (start, end);
    return 0;
}

```

```

(5) #include <stdio.h>

int isprime (int n){
    if (n == 0){
        return 0;
    }
    if (n % 2 != 0)
    {
        return 1;
    } else {
        return 0;
    }
    isprime (n-1);
}

int main () {
    int n;
    scanf ("%d", &n);
    if (isprime (n))
        printf ("prime");
    else {
        printf ("not prime");
    }
    return 0;
}

```

```

(6) #include <stdio.h>
#include <string.h>

void strcpy (char str1[],
             char str2[], int n){
    if (n == 0){
        str2[0] = "\0";
        return;
    }
    str2[n-1] = str1[n-1];
    strcpy (str1, str2, n-1);
}

int main () {
    char str1[100] = "hello this is";
    string s1;
    char str2[100];
    int n = strlen (str1);
    strcpy (str1, str2, n);
    printf ("%s", str2);

    return 0;
}

```

```
① # include <stdio.h>

int findmin (int arr [], int n) {
    if (n == 1) {
        return arr [0];
    }
    int min = findmin (arr, n-1);
    if (min > arr [n-1]) {
        min = arr [n-1];
    }

    return min;
}

int main () {
    int arr [] = { 11, 2, 3, 6, 5 };
    printf ("%d", findmin arr, 5);

    return 0;
}
```